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1.) A 200 MHz left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air on a dielectric medium with ϵ_r = 4 and occupies the region defined by z ≥ 0. (20 points total)

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at z = 0 and t = 0. (5 points)

$\tilde{E}^i = a_0 \hat{e}\, e^{\mp j k_1 u}$

$\boxed{\tilde{E}^i = 5(\hat{x}+j\hat{y})\, e^{-j\frac{4\pi}{3}z}\ V/m}$

$f = 200\ MHz$

$\omega = 2\pi f = 400\pi \times 10^6\ rad/s$

$u = z$

$\hat{e} = LHC = \hat{x} + j\hat{y}$

$k_1 = \frac{\omega\sqrt{\epsilon_{r1}}}{c} = \frac{4\pi}{3}$

$c = 3\times10^8\ m/s$

$a_0 = 5\ V/m$

$\mu_r = \epsilon_{r_1}\ (Air) = 1$

(b) Calculate the reflection and transmission coefficients. (5 points)

$\eta_1\,(Air) = \sqrt{\frac{\mu_{r1}}{\epsilon_{r1}}}\,\eta_0 = \sqrt{\frac{1}{1}}\,(120\pi) = 120\pi$

$\eta_0 = 120\pi$

$\eta_2\,(med.\ 2) = \sqrt{\frac{\mu_{r2}}{\epsilon_{r2}}}\,\eta_0 = \sqrt{\frac{1}{4}}\,(120\pi) = 60\pi$

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \boxed{-\frac{1}{3}}$ (unitless)

$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$ (unitless)

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region z ≤ 0. (5 points)

$$\tilde{E}^r = \Gamma a_0 \hat{e} e^{\pm jk_1 u}$$

$$\tilde{E}^r = -\frac{1}{3}(5)(\hat{x}+j\hat{y})e^{+j\frac{4\pi}{3}z} \text{ V/m}$$

$$\boxed{\tilde{E}^r = -\frac{5}{3}(\hat{x}+j\hat{y})e^{j(4\pi/3)z} \text{ V/m}}$$

$$\tilde{E}^t = \tau a_0 \hat{e} e^{\mp jk_2 u}$$

$$= \frac{2}{3}(5)(\hat{x}+j\hat{y})e^{-j\frac{8\pi}{3}z}$$

$$k_2 = \frac{\omega\sqrt{\varepsilon_{r2}}}{c} = \frac{8\pi}{3}$$

$\varepsilon_{r2} = 4$

$c = 3\times10^8$ m/s

$\omega = 400\pi\times10^6$ rad/s

$$\boxed{\tilde{E}^t = \frac{10}{3}(\hat{x}+j\hat{y})e^{-j\frac{8\pi}{3}z} \text{ V/m}}$$

$$\tilde{E}_T = \tilde{E}^i + \tilde{E}^r$$

$$= 5(\hat{x}+j\hat{y})e^{-j\frac{4\pi}{3}z} - \frac{5}{3}(\hat{x}+j\hat{y})e^{j\left(\frac{4\pi}{3}\right)z}$$

$$\boxed{\tilde{E}_T = 5(\hat{x}+j\hat{y})\left[e^{-j\frac{4\pi}{3}z} - \frac{1}{3}e^{j\frac{4\pi}{3}z}\right] \text{ V/m}}$$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

$$\% P_r = 100|\Gamma|^2 = 100\left|-\frac{1}{3}\right|^2 = \boxed{11.1\%}$$

$$\% P_t = 100|\tau|^2\left(\frac{\eta_1}{\eta_2}\right) = \frac{120\pi}{60\pi} = \boxed{88.9\%}$$

Check work $P = 11.1\% + 88.9\% = 100\%$, energy conserved ✓

2. Repeat the above problem but replace the dielectric medium with a poor conductor characterized by ϵ_r = 2.25, μ_r = 1, and σ_2= 10^−4 S/m (20 points total)

Medium 1: Air $\mu_r = \varepsilon_r = 1$

$\omega = 2\pi f = 400\pi \times 10^6$ rad/s

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at z = 0 and t = 0. (5 points)

$\tilde{E}^i = a_0 \hat{e} e^{\mp j k_1 u}$

$$\boxed{\tilde{E}^i = 5(\hat{x} + j\hat{y}) e^{-j\frac{4}{3}\pi z} \text{ V/m}}$$

$K_1 = \frac{\omega\sqrt{\varepsilon_{r1}}}{c} = \frac{\pi 400 \times 10^6 \sqrt{1}}{3\times 10^8} = \frac{4}{3}\pi$ rad/m

$u = z$ $a_0 = 5$ V/m $\hat{e} = LHC = \hat{x} + j\hat{y}$

(b) Calculate the reflection and transmission coefficients. (5 points)

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

$\eta_1 = \sqrt{\frac{\mu_{r1}}{\varepsilon_{r1}}}\eta_0 = 120\pi\ \Omega$

$\eta_2 = \sqrt{\frac{\mu_{r2}}{\varepsilon_{r2}}}\eta_0 = \sqrt{\frac{1}{2.25}}\,120\pi = 80\pi\ \Omega$

$\Gamma = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \boxed{-\frac{1}{5}}$

$\tau = 1 + \Gamma = 1 - (-\frac{1}{5}) = \boxed{\frac{4}{5}}$

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m
$\omega = 400\pi \times 10^6$ rad/s
$\varepsilon_{r2} = 2.25$
$\mu_{r2} = 1$
$\sigma_2 = 10^{-4}$ S/m
$\eta_0 = 120\pi$
} Medium 2

$\varepsilon_{r1} = 1$
$\mu_{r1} = 1$
} Medim 1 (Air)

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region z ≤ 0. (5 points)

$\eta_0 = 120\pi$ $\quad \Upsilon = \frac{4}{5}$

$a_0 = 5$ V/m $\quad c = 3\times10^8$ m/s

$k_1 = \frac{4}{3}\pi$ rad/m $\quad \omega = 400\pi\times10^6$ rad/s

$u = z$ $\quad \varepsilon_{r_2} = 2.25$

$\Gamma = -\frac{1}{5}$ $\quad \sigma_2 = 10^{-4}$ S/m

$\hat{e} = \hat{x} + j\hat{y}$ $\quad \varepsilon_0 = 8.85\times10^{-12}$ F/m

$$\tilde{E}^r = \Gamma a_0 \hat{e}\, e^{\pm jk_1u}$$

$$= -\frac{1}{5}\,5(\hat{x}+j\hat{y})\, e^{+j\frac{4}{3}\pi z}$$

$$\boxed{\tilde{E}^r = -(\hat{x}+j\hat{y})\, e^{+j\frac{4}{3}\pi z}\ \text{V/m}}$$ reflected wave

$$\tilde{E}^t = \Upsilon a_0 \hat{e}\, e^{\mp jk_2u}$$ but

$$k_2 = \frac{\omega\sqrt{\varepsilon_{r2}}}{c} = \frac{400\pi\times10^6\sqrt{2.25}}{3\times10^8} = \frac{3}{4}\pi$$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_{r_2}\varepsilon_0} = \frac{10^{-4}}{(400\pi\times10^6)(2.25)(8.85\times10^{-12})} = 3.99\times10^{-3}$$

since $\frac{\varepsilon''}{\varepsilon'} << 10^{-2}$, lossless, so $\underbrace{e^{\mp jk_2u} \to e^{\mp\alpha_2 u}\, e^{\mp j\beta_2 u}}$

$$\alpha_2 = \frac{\sigma_2\eta_0}{2\sqrt{\varepsilon_{r2}}} = \frac{(10^{-4})(120\pi)}{2\sqrt{2.25}} = 0.004\pi$$

$$\beta_2 = \frac{\omega\sqrt{\varepsilon_{r2}}}{c} = \frac{400\pi\times10^6\sqrt{2.25}}{3\times10^8} = 2\pi$$

$$\tilde{E}^t = \frac{4}{5}(5)(\hat{x}+j\hat{y})\, e^{-j\,0.004\pi z}\, e^{-j2\pi z}$$

$$\boxed{\tilde{E}^t = 4(\hat{x}+j\hat{y})\, e^{-j0.004\pi z}\, e^{-j2\pi z}\ \text{V/m}}$$ transmitted wave

$$\tilde{E}_{total} = \tilde{E}^i + \tilde{E}^r$$

$$= 5(\hat{x}+j\hat{y})\, e^{-j\frac{4}{3}\pi z} - (\hat{x}+j\hat{y})\, e^{j\frac{4}{3}\pi z}$$

$$\boxed{\tilde{E}_{total} = (\hat{x}+j\hat{y})\left[5e^{-j\frac{4}{3}\pi z} - e^{j\frac{4}{3}\pi z}\right]\ \text{V/m}}$$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

$$\%P_r = 100|\Gamma|^2 = 100\left(-\frac{1}{5}\right)^2 = \boxed{4\%}$$

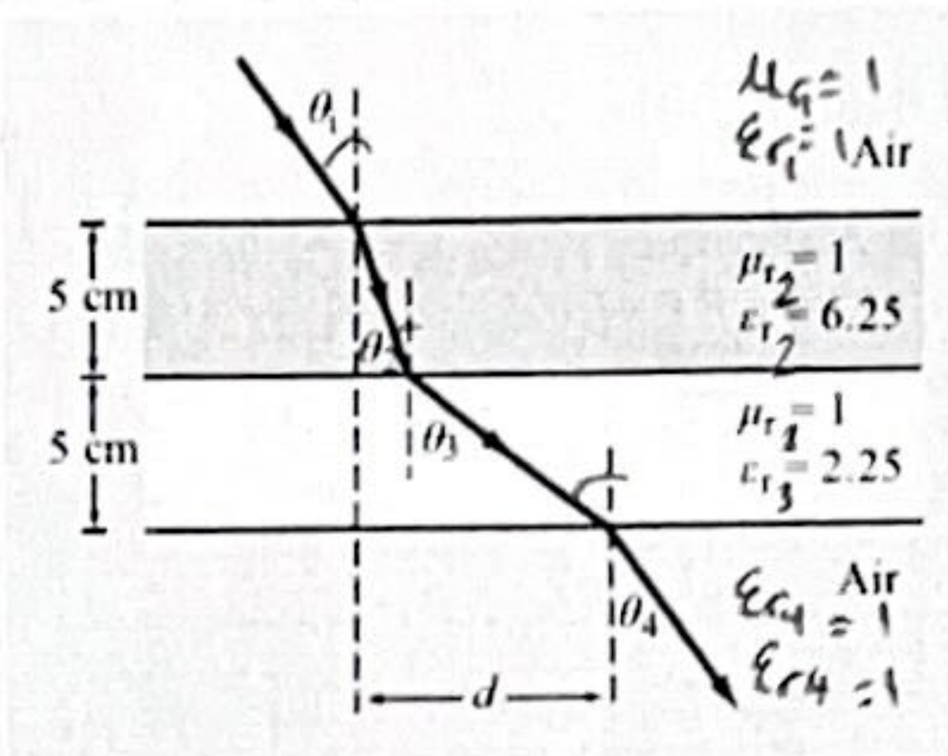
$$\%P_t = 100|\Upsilon|^2\frac{\eta_1}{\eta_2} = 100\left(\frac{4}{5}\right)^2\frac{120\pi}{80\pi} = \boxed{96\%}$$

3. A parallel-polarized plane wave is incident from air at an angle θ_i = 30∘ onto a pair of dielectric layers, as shown above. (10 points total)

$\theta_1 = 30^\circ$

$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\varepsilon_{ri}}{\varepsilon_{ri+1}}}$

$\theta_{i+1} = \sin^{-1}\left(\sin(\theta_i) \sqrt{\frac{\varepsilon_{ri}}{\varepsilon_{ri+1}}}\right)$



(a) Determine the angles of transmission θ_2, θ_3, and θ_4. (5 points)

$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}}$

$\sin \theta_2 = \sin(30^\circ) \sqrt{\frac{1}{6.25}}$

$\boxed{\theta_2 = 11.54^\circ}$

$\theta_3 = \sin^{-1}\left(\sin(\theta_2) \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r3}}}\right)$

$= \sin^{-1}\left[\sin(11.54^\circ) \sqrt{\frac{6.25}{2.25}}\right]$

$\boxed{\theta_3 = 19.47^\circ}$

$\theta_4 = \sin^{-1}\left[\sin(\theta_3) \sqrt{\frac{\varepsilon_{r3}}{\varepsilon_{r4}}}\right]$

$= \sin^{-1}\left[\sin(19.47^\circ) \sqrt{\frac{2.25}{1}}\right]$

$\boxed{\theta_4 = 30.00^\circ}$

(b) Determine the lateral distance d. (5 points)

$t_2 = t_3 = 0.05 m$

$d = \sum_{i=1}^{N} t_i \tan \theta_i$

$= t_2 \tan(\theta_2) + t_3 \tan(\theta_3)$

$= 0.05 \tan(11.54^\circ) + 0.05 \tan(19.47^\circ)$

$\boxed{d = 0.0279 \text{ m}}$

4. A TE wave propagating in a dielectricfilled waveguide of unknown permittivity has dimensions a = 5 cm and b = 3 cm. If the x componentof its electric field is given by

$= 0.05m$ $= 0.03m$

$$E_x = -36\cos(\underbrace{40\pi x}_{\frac{m\pi x}{a}})\sin(\underbrace{100\pi y}_{\frac{n\pi x}{b}})\sin\left(\underbrace{(2.4\pi\times10^{10})}_{\omega}t - \underbrace{52.9\pi}_{\beta}z\right)\ \text{(V/m)},$$

determine: (20 points total)

(a) the mode number, (5 points)

TE_{mn} (Given from queston "TE Wave")

$$\frac{m\cancel{\pi}\cancel{x}}{a} = 40\cancel{\pi}\cancel{x}$$

$$\frac{m}{(0.05)} = 40$$

$$m = 2$$

$$\frac{n\cancel{\pi}\cancel{y}}{b} = 100\cancel{\pi}\cancel{y}$$

$$\frac{n}{0.03} = 100$$

$$n = 3$$

$TE_{mn} = \boxed{TE_{23}}$

(b) ϵ_r of the material in the guide, (5 points)

Given: $\beta = 52.9\pi$, $\omega = 2.4\pi\times10^{10}$ rad/s, $c = 3\times10^8$ m/s

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$$

$$= \frac{(3\times10^8)^2}{(2.4\pi\times10^{10})^2}\left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2\right]$$

$$\boxed{\varepsilon_r = 2.25}$$

(c) the cutoff frequency, (5 points)

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad \text{(Hz)}$$

$$\frac{m}{a} = 40$$

$$\frac{n}{b} = 100$$

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3\times10^8}{\sqrt{2.25}} = 200\times10^6$$

$$f_{23} = \frac{200\times10^6}{2}\sqrt{40^2 + 100^2} \quad \text{Hz}$$

$$= \boxed{10.77 \text{ GHz}}$$

(d) and the expression for H̃_y. (5 points)

$$\tilde{H}_y = \frac{\tilde{E}_x}{Z_{TE}}$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{2.25}} = 80\pi$$

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi} = \frac{2.4\pi\times10^{10}}{2\pi} = 12\times10^9 \text{ Hz}$$

$$Z_{TE} = \frac{\eta}{\sqrt{1-(f_c/f)^2}}$$

$$= \frac{80\pi}{\sqrt{1-(10.77\times10^9/12\times10^9)^2}}$$

$$= 569.9 \ \Omega$$

$$\tilde{H}_y = \frac{-36}{569.9}\cos($$

$$\boxed{\tilde{H}_y = -0.06317\cos(40\pi x)\sin(100\pi y)\sin(2.4\pi\times10^{10}t - 52.9\pi z) \ \text{V/m}}$$

5. A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with a = 3 cm and b = 2 cm. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end? (8 points)

f_{mn} = $f_c = 9.5 \times 10^9$ Hz $a = 0.03$ m $b = 0.02$ m

$$f_{mn} = \frac{u_{po}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3\times10^8}{2}\sqrt{}$$

$$u_{po} = \frac{c}{\sqrt{\varepsilon_r}} = 3\times10^8 \text{ m/s}$$

Mode	$f_c = f_{mn}$	$u_g = u_{po}\sqrt{1-(f_{mn}/f)^2}$	$t = L/u_g$
TE_{10}	f_{10} $\frac{u_{po}}{2a} = \frac{3\times10^8}{2(0.03)}$ $= 5\text{GHz} << 9.5G$ Excited	$= 3\times10^8\sqrt{1-\left(\frac{5G}{9.5G}\right)^2}$ $= 255.1\times10^6$ m/s	$\frac{100}{255.1\times10^6}$ $= \boxed{0.392\,\mu s}$
TE_{01}	$f_{01} = \frac{u_{po}}{2b}$ $= \frac{3\times10^8}{2(0.02)}$ $= 7.5G << 9.5G$ Excited	$= 3\times10^8\sqrt{1-\left(\frac{7.5G}{9.5G}\right)^2}$ $= 184.1\times10^6$ m/s	$= \frac{100}{184.1\times10^6}$ $= \boxed{543.1\,nS}$
TE_{20}	$f_{20} = \frac{u_{po}}{a}$ $= \frac{3\times10^8}{0.03}$ $= 10$ GHz $> 9.5G$ Not excited	N/A Not Excited	NA
TE_{11} TM_{11}	$f_{11} = \frac{u_{po}}{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$ $= \frac{3\times10^8}{2}\sqrt{\frac{1}{0.03^2}+\frac{1}{0.02^2}}$ $= 9.0$ GHz $< 9.5G$ Excited	$u_g = u_{po}\sqrt{1-\left(\frac{9.01G}{9.5G}\right)^2}$ $= 95.10\times10^6$ m/s	$= \frac{100}{95.10\times10^6}$ $= \boxed{1.02\,\mu s}$